



Selection, grouping, and assignment policies with learning-by-doing and knowledge transfer



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ABSTRACT

This paper investigates the allocation of workers to tasks based on individual learning characteristics in order to improve system throughput. We model worker productivity to include skill knowledge obtained by learning-by-doing and learning-by-transfer. The allocation problem is composed of the selection of workers from a pool, grouping workers based on individual characteristics, and assignment of groups to tasks. We examine several related heuristic approaches for each phase with interest in identifying useful measures and policies, as well as interactions among these policies. The performance of each heuristic is compared to a baseline policy and to an upper bound obtained by solving a non-linear math-programming problem. Results are examined in the context of both parallel and serial systems. Current results demonstrate that heuristics based on estimated output, ability to learn from other workers, and asymptotic steady state productivity are useful for selecting, grouping and assigning workers to tasks, and these policies perform well with respect to objective gap relative to an upper bound.

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1. Introduction

The combination of short product life cycles, intense competition, and market globalization has created increased pressure on manufacturing and service industries. Many of these market challenges have forced organizations to migrate toward greater agility and flexibility. Additionally, innovation has become a key factor in increasing market share and as a consequence, processes involved in the production of goods and services exist in a state of perpetual change. For organizations, it has become critical to learn, adapt and integrate faster than the competition to survive economically (Kapp, 1999). At the facility level, the ability of an organization to learn faster relies on the ability of individual workers to learn and transfer skills rapidly. In addition to having the most appropriate staffing when organizing production, managers might further consider strategies that enhance the transfer of knowledge among the workforce in order to hasten the learning process.

Learning is considered to be ubiquitous in contemporary organizations. Nonetheless, questions of how learning, knowledge, cognition and other behavioral characteristics affect organizations remain open. Indeed, while some have suggested that organizational learning is the sum of what individuals in the organization

learn (e.g., Kim, 1993; Simon, 1991), others have argued that it is a result of the collective activities, ideas, systems, processes, and structures of the organization (Levitt & March, 1988; March, 1991). The debate has spurred the interest in learning as a concept from a variety of perspectives. For instance, Gherardi and Nicolini (2001) discussed the sociological foundations of organizational learning. Also, Yelle (1979) modeled learning and organizational learning, including technological learning. The question of why some organizations learn faster than their competitors was described by Abernathy, Clark, and Kantorow (1981) and Fine (1986). Zangwill and Kantor (1998) suggested a learning-curve model that accounts for learning through continuous improvement efforts. Since organizational learning occurs through and among people, attention has been given to the exploration of how learning takes place and how it is facilitated among individuals and within groups. Individual learning processes have been studied and modeled by psychologists (e.g., Mazur & Hastie, 1978). Engineers have modeled learning as it relates to manufacturing costs (Yelle, 1979), and setup time learning (Pratsini, Camm, & Raturi, 1993). Further, the importance of heterogeneity within a workforce was highlighted by Shafer, Nembhard, and Uzumeri (2001). Human learning characteristics can be represented by various individual-based parameters such as steady state productivity rates, previous expertise, and learning and forgetting rates. Teyarachakul, Chand, and Ward (2011) and Nembhard and Osothsilp (2001) respectively,

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present a survey, and a comparison of several learning and forgetting models. Since organizations may benefit from knowledge acquired by others, we will consider another type of learning in addition to learning by doing, that of learning from others' experience.

At the individual level, transfer of knowledge may take place when working closely with fellow workers who are doing the same or related jobs. The ability of individuals to use knowledge already accumulated by their colleagues has been described as a means to improve the rate of knowledge transfer within organizations. Research in this area provides several related explanations for why working together improves performance, which in turn are based on varied mechanisms. One is the ability of workers to coordinate their activities. When studying group dynamics, the importance of knowing "who knows what" has been highlighted. Faraj and Sproull (2000), Larson, Christensen, Abbott, and Franz (1996), Lewis (2003), Moreland, Argote, and Krishnan (1996), and Wegner (1986, 1995) suggest that teams composed of individuals who have experience working together are more accurate and have a shared sense of who knows what on a team. Liang, Moreland, and Argote (1995) discuss the importance of this, arguing that it promotes the development of a more effective division of labor. Researchers studying shared work experience have emphasized the importance of transaction partners learning and how to govern their interactions. Uzzi (1996) suggests that multiple exchanges increase the likelihood of team trust. Uzzi and Lancaster (2003) extend the idea by discussing how trust can promote the exchange of "private" knowledge and information. The various explanations for why experience working together improves coordination tend to complement one another. Knowing "who knows what" is critical for defining roles and responsibilities inside the team and for assigning the most competent person to each role. However, this is only one half of effective coordination. The other half turns on the development of relationship-specific heuristics that enhance how well people performing distinct roles interact with one another.

In collaborative work environments, the transfer of knowledge to a new hire may occur as a result of being in the presence of other people who are at a higher level of competence obtained via previous experience. Learning is often a slow process where the time required to obtain significant learning may vary from a few days to several months. Workers learn and forget based on how often they perform tasks, where the existence and importance of learning effects on productivity have been known for some time (e.g., Cochran, 1960; Conway & Schultz, 1959; Venezia, 1985). The impact of learning on system performance has led a number of researchers to consider workforce skills in the worker-task assignment problem. Warner, Needy, and Bidanda (1997) suggested assigning workers to machine cells based on their human and technological skills. Similarly, Bhaskar and Srinivasan (1997) presented math-programming models for static and dynamic worker assignment problems in cellular manufacturing, developing and testing the performance of proposed heuristics when minimizing makespan. Norman, Tharmmaphornphilas, Needy, Bidanda, and Warner (2002) developed a team-based model for worker assignment in manufacturing cells with both technical and human skills, demonstrating that when skills are explicitly taken into account in both training plans and assignment strategies, the performance of the cell will improve significantly. Nembhard (2001) proposed a heuristic approach for selecting workers for tasks based on individual learning rates, and Nembhard and Osothsilip (2005) presented an approach for selecting workers for tasks of varying complexity based on individual learning and forgetting characteristics in order to improve system productivity. Also, Nembhard and Bente fouet (2012) established an optimal schedule when assigning workers to tasks with general learning and forgetting, and delineated opti-

mal flow line conditions in the context of human performance variability. To date, the current study is the first to propose and consider learning by doing and knowledge transfer in the worker-task selection, grouping and assignment problem.

The remainder of the paper is organized as follows. Section 2 is devoted to the presentation of the learning model while section 3 focuses on the methodology. In Section 4, we summarize the results of the simulation design and discuss the outcome in section 5.

2. A model for learning-by-doing and knowledge transfer

While learning by doing has been quantified and modeled in various ways, the literature on learning by transfer of knowledge is relatively scarce. Much of the work in this area has been descriptive, rather than model based. For example, Salomon and Globerson (1987), and Lim and Johnson (2002) examine issues of training and knowledge transfer mechanisms. However, models of learning, such as the Mazur and Hastie (1978) model in Eq. (1), describe a hyperbolic function that behaves well for both mechanical and cognitive learning by doing. The model was shown to provide the best fit among eleven models for manual skill learning (Nembhard and Uzumeri, 2000). Their multi-criteria comparison considered the average model fit among individuals, the variance of the fit, the number of parameters, and the ability to capture episodes of negative behavior.

$$y = k \left(\frac{x + p}{x + p + r} \right) \quad (1)$$

$y, k, p, x \geq 0$ and $p + r > 0$

In Eq. (1), y is a measure of work performance, x is the amount of cumulative work in units of time or trials, and p represents the individual's accumulated prior expertise for that task in the same units as x . The parameter k estimates the asymptotic limit of y . That is, in each model, k estimates the asymptotic steady-state performance level that can be expected when all learning has been completed. Parameter r is the cumulative production required to attain an output level of $k/2$. The three parameters k , p , and r have the advantage to have straightforward interpretation which should aid in the process of transferring knowledge of workplace performance into the managerial decision making process.

In this paper, we focus on learning and propose a modification to Eq. (1) that considers learning-by-transfer of knowledge from others in the workplace. That is, when an operator is working in a team environment, not only does s/he learn from direct first-hand experience but also from potential transfer of knowledge from others in the group. We represent this process by modeling knowledge as the sum of direct experience, x , measured by the number of units of work completed, and indirect experience measured by the experience of others on the team, T , scaled by an individual-based parameter, θ , that represents individual ability to learn via transfer. Thus, Eq. (2) modifies Eq. (1), to account for potential effective experience, acquired by transfer and/or direct experience.

$$y_x = k \left(\frac{\theta * T + x + p}{\theta * T + x + p + r} \right) \quad (2)$$

With the additional parameters: θ is the transfer factor, and T the total pool of knowledge (cumulative work of all other operators). The parameter θ quantifies the percentage of learning that is transferred from other workers, and can represent both negative and positive transfer of knowledge. Alternately, if the transfer parameter θ is bounded by 0 and 1, only positive transfer is estimated. Another advantage of the model is that each model parameter, k , p , r and θ has a useful interpretation, and inferences on parameters and relationships between parameters can aid in managerial

decision making. We note that more of each parameter is generally preferred, except for r , where smaller values are preferred; we will later use $1/r$, in candidate policies. To implement the model, we describe the full explication of the model in Eq. (3), which accounts for each combination of worker, i , and task, j , and the index of the production period t .

$$y_{ijt} = k_{ij} \left(\frac{\theta_{ij} * T_{ijt} + \sum_{k=1}^t x_{ijk} + p_{ij}}{\theta_{ij} * T_{ijt} + \sum_{k=1}^t x_{ijk} + p_{ij} + r_{ij}} \right) \quad (3)$$

The direct cumulative experience is represented by $\sum_{k=1}^t x_{ijk}$, which sums the number instances the worker i has been assigned to task j up to and including period t . The team experience term, T_{ijt} , estimates the productivity at time t of other workers performing tasks similar to task j and is more formally defined in [Appendix A](#).

3. Methodology

This study aims to address important questions managers face in the workplace when organizing production. We conduct a range of simulations acknowledging that decision-making can occur at discrete times when allocating workers to tasks. The simulations will be informed by empirical work characteristics based on the model in Eq. (3), with data as described below in Section 3.1. We organize the scheduling problem into three consecutive stages: (1) a *selection* stage that addresses the question of who; based on a ranking method, (2) a *grouping* stage wherein the selected subset of workers can have the advantage of knowledge transfer, and (3) an *assignment* stage, that addresses the question which tasks are to be allocated to each member of a group.

3.1. Data

The methodology involves investigating production systems, wherein we simulate worker performance using the productivity model described in Eqn. (3) and secondary data from the literature. The model and data match those from a final test and inspection station at car radios production lines ([Shafer et al., 2001](#)). In Shafer et al., time stamp data were collected for 176,000 items inspected by 75 workers. The production lines included a combination of machine-paced and human-paced activities. The machine-paced activities were computer controlled and exhibited negligible variability. The model in Eq. (3) was fit to each of the 75 workers' data with an average R^2 of 96%, and standard deviation of 2% using non-linear least squares curve fitting. The fitting process yields a 4-tuple of model parameter values (k, p, r, θ) for each worker. To illustrate, a sample of the fitted model parameters includes three operators with respective 4-tuples with natural logarithms of the first three parameters in order to yield normal univariate distributions. That is, sample 4-tuples ($\log(k), \log(p), \log(r), \theta$) include $A(1.39, 2.46, 1.59, 0.0067)$, $B(1.62, 3.03, 3.12, 0)$ and $C(1.48, 0, 4.17, 0.16)$. For instance, operator B has a relatively high steady state productivity rate, k , yet is unreactive to transfer of learning from the group of co-workers, and $\theta_B = 0$. Operators A and C each had some transfer of learning from their co-workers, with operator C more receptive of the two in learning from co-workers' cumulative experience.

These data, from the full set of 75 operators, are then used to inform the simulation of worker performance behavior within the systems and policies that we examine. [Table 1](#) presents the log means of parameters (k, p, r) for the workers, with logs taken in order to yield normal univariate distributions. The parameter θ has a mean of 0.664 and a variance of 0.409, and is normally distributed without transformation.

In the simulation design described below, a worker's performance on a task is estimated using the productivity model

Table 1

Learning parameter means, variance–covariance matrix.

$\mu = \begin{bmatrix} 1.448 \\ 1.963 \\ 2.0446 \end{bmatrix}$			$\Sigma = \begin{bmatrix} \log k & \log p & \log r \\ 0.0045 & 0.0167 & 0.0191 \\ 0.0167 & 0.4621 & 0.2443 \\ 0.0191 & 0.2443 & 0.1905 \end{bmatrix}$		
			$\log k$	$\log p$	$\log r$

described in Eq. (3). We note that the above-described parameterization forms a MultiVariate Normal (MVN) distribution, which we use throughout the simulations.

3.2. Simulation design

The production setting considered has three types of machines (numbered 1–3), each with three identical instances of each machine for a total of nine machines, to be staffed by nine selected workers. This system size was chosen to have a manageable, yet potentially rich set of dynamics and interactions to investigate. We note that transfer may occur among workers staffing machines they share in common, but not for different machine types. For clarity, we assume that for each machine type, workers have common learning and transfer parameters. However, experience and performance on each are accounted for individually. Both parallel and serial system configurations are of interest with their respective configurations given in [Fig. 1](#) and [Fig. 2](#).

3.3. Selection stage

We first select nine workers from a pool of 45 workers sampled from the MVN, these nine will later be grouped and subsequently assigned to tasks. We examine the question of how to select a workforce for a production system considering several ranking methods, wherein the top nine ranked workers are retained for the grouping and assignment stages. Rankings are based on either, the highest overall (*MaxiMax*) or the highest minimum (*MaxiMin*) value with regard to each parameter (learning parameter, transfer parameter or expected output) among all three types of tasks, resulting in 10 considered selection criteria. That is, the 10 criteria include the two objective methods (*maximax* and *maximin*) applied to each of the five parameters $k, p, 1/r, \theta$, and O . That is, k represents the steady state production rate from the learning model, and p is the prior level of expertise as imputed from the empirical data. The reciprocal of the learning parameter is used in order to make all criteria have the same “more is better” characteristic. The parameter θ , is the individual learning transfer parameter. Lastly, O is calculated using Eq. (1) to represent the *expected output* when performing a task during the same duration as the experiment providing a relative estimate of individual production over the time horizon. The rationale for including this calculated parameter stems from combining the four parameters into a single relative measure of worker ability.

3.4. Grouping stage

In order to examine the potential importance of transfer of knowledge when workers perform similar tasks within a common environment, the selected workers are organized into groups. The nine selected workers from the selection stage are grouped into teams of three, corresponding to three machine types. We consider several strategies for group formation, based on within-team variances among workers with respect to the learning and transfer parameters. Specifically, we consider maximal variance and minimal variance among each of the parameters and the predicted system output. For example, for the learning parameter r , within-team

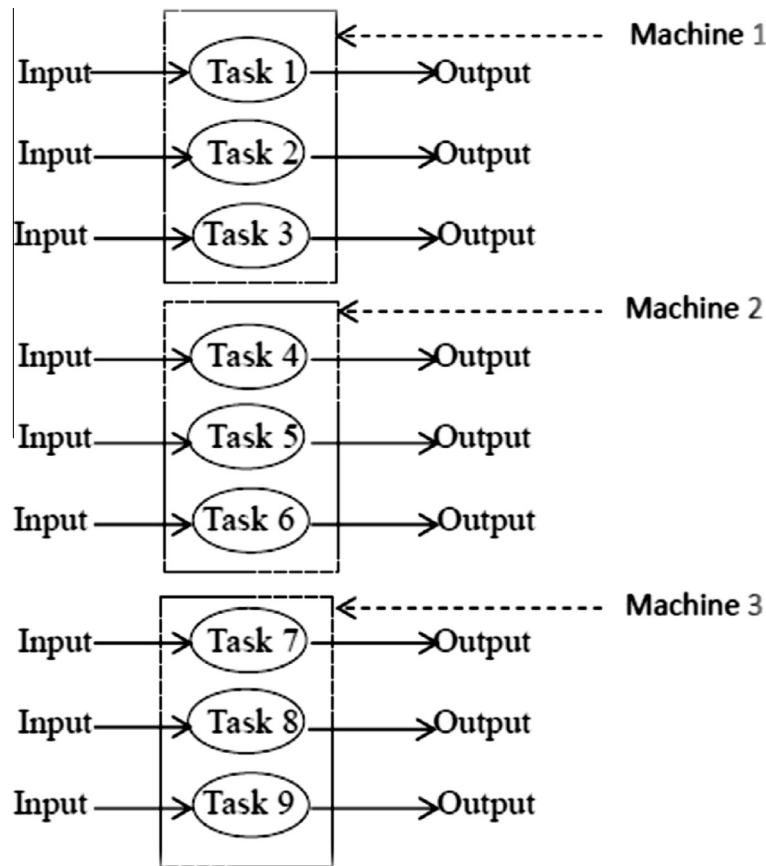


Fig. 1. Parallel system configuration (9 tasks).

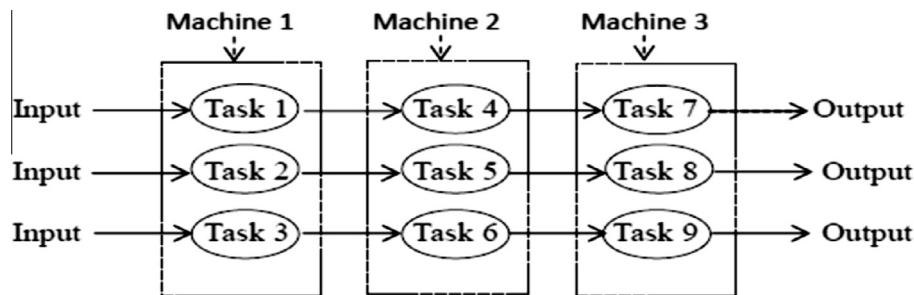


Fig. 2. Serial system configuration (9 tasks).

variance is as the sum of the square variations of r_i , among each worker i , grouped to a given team.

3.5. Assignment stage

The assignment stage addresses the question of which subset of tasks should be performed by each team of workers. Teams of three workers are assigned to a set of replicated tasks forming a class and will only perform tasks within that class. The assignment strategies are based on assigning teams to tasks that have either the highest (*MaxiMax* assignment) or the lowest (*Maximin* assignment) parameter value. In a manner similar to that for the other stages, both objectives are applied to each of the four parameters, plus the expected output, yielding 10 criteria to be considered. The process starts by assigning the team with the worker that has either, the

highest or the lowest parameter or expected output among all three teams. This process is replicated for each team.

The full matrix of strategies for the three stages (i.e., selection, grouping, and assignment) is summarized in Table 2. A solution

Table 2
Matrix of policies by stage and parameter.

Parameters	Stage	Objective
$(O, k, p, 1/r, \theta)$	Selection	Maximax
		Maximin
	Grouping	Max Var
		Min Var
	Assignment	Maximax
		Maximin

O is the expected output with the productivity model described in Eq. (1).

to the allocation problem is then a combination of policies at these stages. The policies considered in this paper are candidate heuristic policies selected to yield general insights, recognizing that exact optimizing at any one of these stages is in itself a computationally intensive combinatorial problem (Bentefouet and Nembhard, 2013). Given the 10 policies for each of the three stages, we will examine a total of one thousand feasible ways to select, group and assign workers. We will then compare strategies within stages, and as combinations across the stages. For example, given a population of 45 workers, we apply selection methods based on individual worker rankings, then for each selection method, grouping strategies are implemented and for each grouping strategy, the assignment approaches are simulated. We consider 25 replications for which a new population of 45 workers is sampled, recognizing that in general organizations do not have complete control their available workforce.

While the *Maximax* policies can be viewed as “greedy” strategies, the *Maximin* approaches may tend to have more balanced worker skills. Further, these provide relative bounds on what straightforward policies may yield, providing potential insight to useful parameters and policies for decision making regarding workforce allocation. Among the questions we examine are, the relative importance of each stage in the allocation problem, and the relevancy of the parameters in the three stages of the allocation problem. Fig. 3 illustrates the overall flow over the three-stage process diagrammatically.

We note that while it is quite common in practice for workforce selection, grouping, and assignment to be conducted in a distinctly separate, uncoordinated manner, there is the additional local optimality issue, wherein ultimately the three stages should be optimized as a single, albeit more complex problem. To evaluate the quality of the implemented policies, their performances are compared to a baseline random allocation of workers to tasks and to an upper bound solution obtained by solving a non-linear scheduling problem where we relax the grouping constraint and the exclusive task type restriction. The mixed integer nonlinear program (MINLP) is presented in Appendix A. The baseline random allocation of worker to tasks is obtained by first randomly selecting 9 workers from the 45 sampled workers, then we arbitrarily group them in exclusive teams of 3 workers, and finally we randomly assign teams of workers to each task type. The following section presents the results of the experiments.

4. Results and discussion

The ANOVA results for evaluating the three stages are shown in Table 3 with dependent variable being the system output. We remark that the *system output* represents the output resulting from applying the choices made at each decision stage. This is distinct from the *expected output*, O , which we defined to be simply the cumulative relative output for a given worker over the time horizon. All three stage factors, selection, grouping, and assignment are significant in both parallel and serial production scenarios. The mean square statistics reveal that the selection stage accounts for most of the variability in both parallel and serial systems, followed by the assignment stage, and lastly the grouping stage. In comparison to the other stages, the grouping stage contributes a small portion of the total variation. The implications of this clearly indicate that selecting the most appropriate workforce is critical. Operational decisions such as task assignment, and grouping in teams is also important, but have a lesser impact on system efficiencies overall than selection. Stated differently, the within group level of heterogeneity, has a relatively low impact on system throughput. While the human resource selection literature is replete with examples illustrating the importance of worker selection (e.g., Wilk & Cappelli, 2003), we remark that the current results also clearly indicate that assignment and grouping are also significant and important, once the selection process is completed.

With the exception of the selection – grouping interaction for parallel systems, all the pairwise interactions are significant. The three-level interaction between selection, grouping, and assignment is not significant in either scenario. Thus, generally, it may not be necessary to evaluate all three stages at once, in order to obtain the most preferred solutions. For example, optimizing the assignment is dependent on how the selection was performed, etc.

Table 4 and Table 5 present the top-ten 3-stage policies (i.e., a combination of a selection, grouping, and assignment of the 1000 treatments) for parallel and serial systems, respectively. The gap is determined by calculating the percentage difference from the upper bound obtained by solving the allocation problem presented in Appendix A.

Both parallel and serial top ten 3-stage policies perform similarly to one another (roughly 0.5% difference between rank 1 and rank 10 policies for each). The parallel system policies are highly efficient, with 1–1.5% gaps. The serial system 3-stage policies were

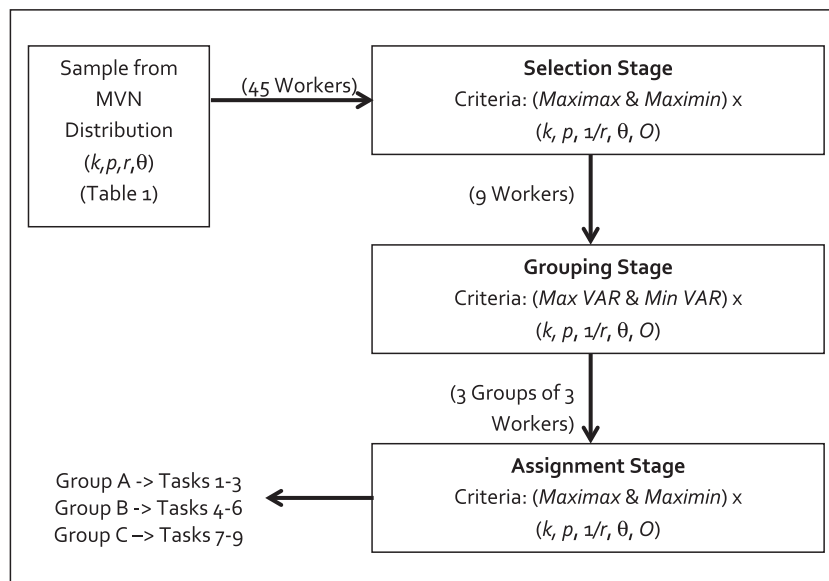


Fig. 3. Diagram of three stages.

Table 3

ANOVA table for simulation treatments (dependent variable: system output).

Source	d.f.	Parallel system			Serial system		
		Mean Sq	F-value	Pr > F	Mean Sq	F-value	Pr > F
Selection	9	721716.7	3092.05	0.000*	77058.3	144.6	0.000*
Grouping	9	2294.6	9.83	0.000*	2780.5	5.22	0.000*
Assignment	9	154441.4	661.67	0.000*	29811.2	55.94	0.000*
Selection*Grouping	81	197.4	0.85	0.837	1379.9	2.59	0.000*
Selection*Assignment	81	1170.8	5.02	0.000*	1223.1	2.3	0.000*
Grouping*Assignment	81	8050.9	34.49	0.000*	2743	5.15	0.000*
Selection*Grouping*Assignment	729	241.2	1.03	0.2625	398.3	0.75	1.0000

* Significance at 5% level.

Table 4

Top ten 3-stage policies: parallel system.

Rank	Selection policy	Grouping policy	Assignment policy	Mean throughput	Mean gap % (std dev. %)
1	Maximax k	Min variance k	Maximax k	808.89 (3.681)	1.15 (1.03)
2	Maximax k	Min variance p	Maximax θ	808.89 (3.681)	1.15 (1.03)
3	Maximax k	Min variance k	Maximax θ	807.77 (3.602)	1.15 (1.06)
4	Maximax k	Min variance $1/r$	Maximax k	807.77 (3.602)	1.15 (1.06)
5	Maximax k	Min variance $1/r$	Maximax θ	807.77 (3.602)	1.15 (1.06)
6	Maximax k	Min variance p	Maximax k	806.76 (4.124)	1.16 (1.06)
7	Maximax k	Min variance θ	Maximax θ	806.56 (3.818)	1.16 (0.95)
8	Maximax k	Min variance θ	Maximax k	804.88 (6.148)	1.18 (1.03)
9	Maximax k	Min variance O	Maximax p	803.02 (14.025)	1.54 (1.36)
10	Maximax θ	Min variance O	Maximax p	803.00 (13.715)	1.56 (1.43)

Table 5

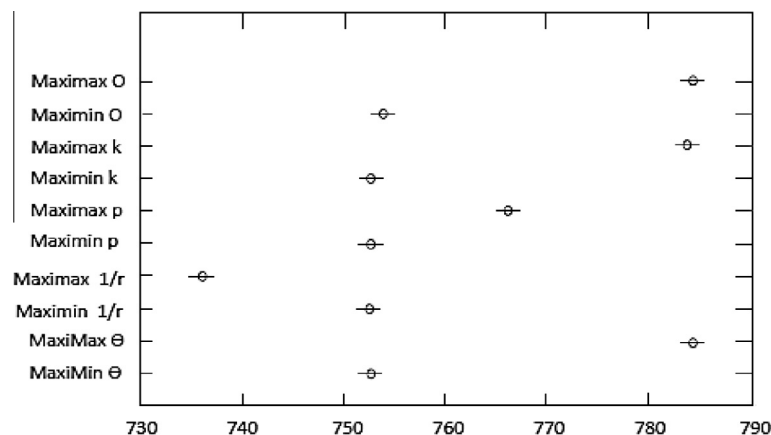
Top ten 3-stage policies: serial system.

Rank	Selection policy	Grouping policy	Assignment policy	Mean throughput	Mean gap % (std dev. %)
1	Maximax θ	Min variance $1/r$	Maximax O	259.07 (7.670)	6.22 (3.14)
2	Maximax θ	Min variance k	Maximax θ	258.02 (7.691)	6.28 (3.14)
3	Maximax θ	Min variance k	Maximax k	258.63 (7.812)	6.38 (3.29)
4	Maximax k	Min variance O	Maximax O	258.62 (7.409)	6.38 (3.03)
5	Maximax θ	Max variance p	Maximax k	258.56 (6.421)	6.42 (2.43)
6	Maximax k	Min variance k	Maximax θ	258.48 (7.275)	6.44 (3.02)
7	Maximax θ	Max variance p	Maximax θ	258.40 (6.437)	6.47 (2.42)
8	Maximax k	Min variance p	Maximax θ	258.11 (7.216)	6.57 (2.94)
9	Maximax O	Min variance k	Maximax k	257.78 (6.994)	6.69 (2.98)
10	Maximax k	Min variance O	Maximax $1/r$	257.63 (12.522)	6.75 (4.73)

somewhat less efficient, with between 6% and 7% gaps. Broadly speaking, selection strategies based on *maximax k* and *maximax θ* performed well. We generated random policies at each stage as points of comparison, yielding mean throughput values of 752 units, and 192 units for the parallel and serial systems, respectively. Comparing these values to our 3-stage policies, gives clear

indication that these policies make a substantive difference from uninformed workforce allocation decision-making.

Figs. 4–9 present the individual-stage mean effects for the two production system types. Figs. 4 and 5 indicate that in parallel and serial systems, with the exception of the parameter r , when selecting workers, the *maximax* policies lead to higher system perfor-

**Fig. 4.** Mean effect of selection policy – Parallel system.

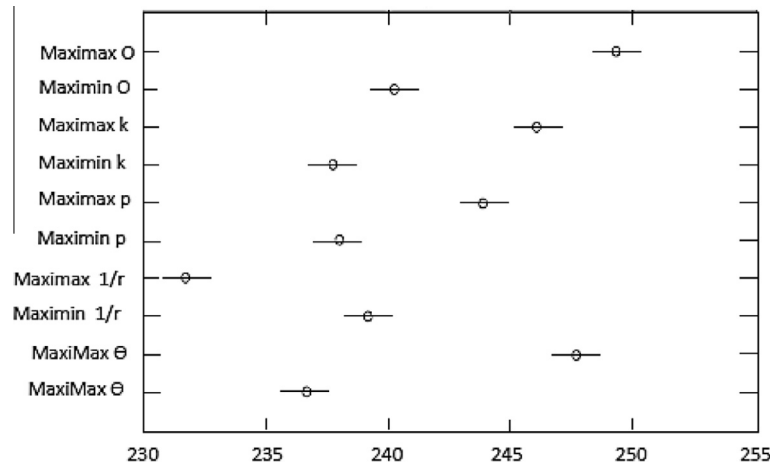


Fig. 5. Mean effect of selection policy – Serial system.

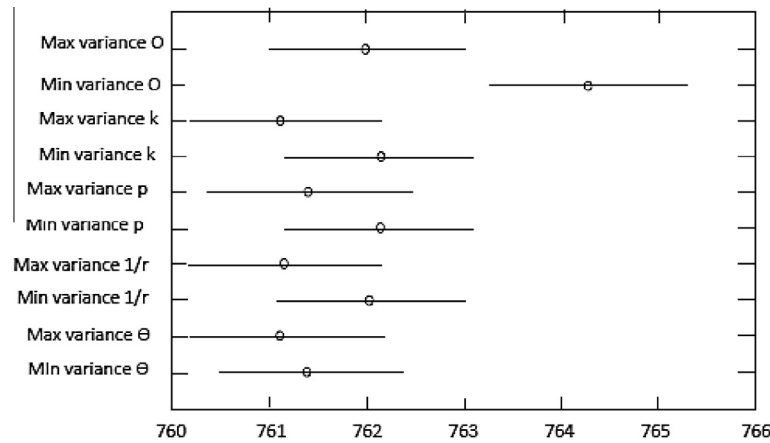


Fig. 6. Mean effect of grouping policy – Parallel system.

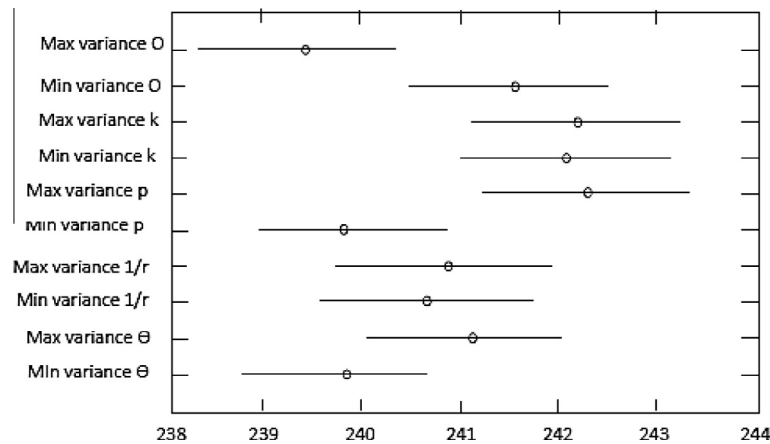


Fig. 7. Mean effect of grouping policy – Serial system.

mance than the corresponding *maximin* policies. *Maximax* policies for output (O), asymptotic steady state (k), and previous expertise level (p) are significantly different in the serial system yet not significantly different in the parallel production scenario. Among all selection policies, the *maximax 1/r* policy leads to the least production performance.

Among the selection stage policies, the *maximax* policies with regard to parameter O , k and, θ dominate the top 30 best policies

(see Appendix B). The parameter O is obtained by calculating with Eq. (1) the output when performing a task during the same duration as the experiment hence it gives a good estimate of the worker production at the end of the simulation. The performance of the policy with regard to k can be explained by the importance of the parameter in the hyperbolic model. Nembhard and Shafer (2007) study this hyperbolic model in detail. The experiment duration is equivalent to a production length of one year that allows the

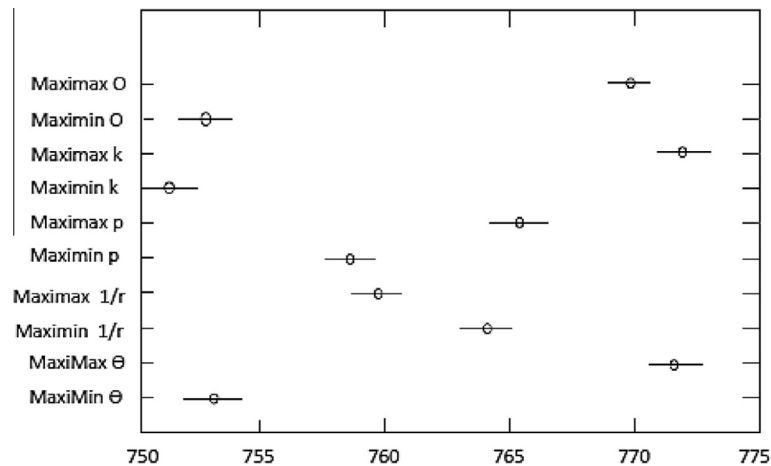


Fig. 8. Mean effect of assignment policy – Parallel system.

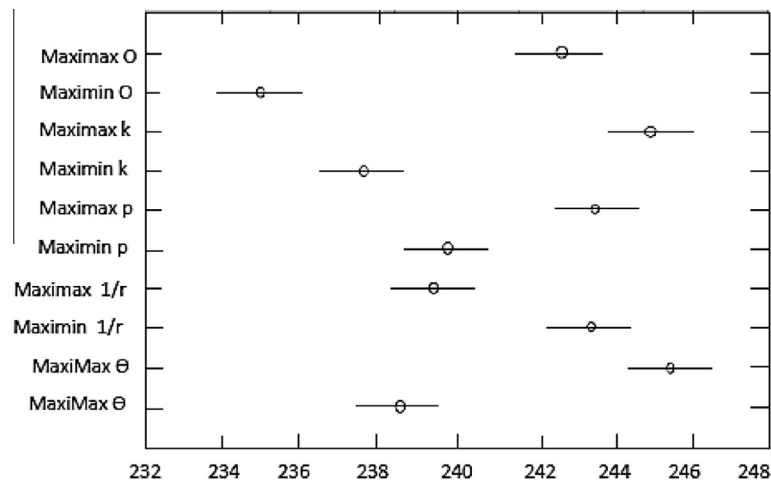


Fig. 9. Mean effect of assignment policy – Serial system.

selected workers to reach their asymptotic steady state productivity. When the steady state productivity is reached, the productivity model converges to the value of parameter k and the performance of policies based on expected output, O , become similar to those based on k . The parameter transfer parameter, θ , represents the amount of knowledge that can be transferred to a worker from other workers performing a similar task. The transferred knowledge is manifested by an increment in the productivity rate, increment proportional to the production of workers performing similar tasks. The performance of *maximax* θ selection policy suggests that selecting workers with regard to the ability to learn from co-workers is as important as hiring workers based on their other learning-by-doing parameters. Thus, at the recruitment stage, attention might be given to understand worker capabilities to work and learn from others.

Figs. 6 and 7 illustrate the impact of the implemented policies when grouping workers into teams. In the parallel system, when forming teams, the policies that minimize the variance among the workers are on average preferable to the ones based on heterogeneity. However the differences with respect to system throughput are only significant when based on the expected output, O . That is, one must calculate the estimate, O , from the four parameters, with the difference between having either dissimilar or similar workers on any individual parameter not being significant. At the grouping stage, the relative difference in system throughput

between the least and the most effective policy among those implemented is less than 0.7 percent in the parallel production scenario and less than 2 percent in the serial production scenario. Thus, grouping workers by similarities or dissimilarities with regard to individual parameter does not have a large impact on the system throughput relative to the selection and assignment stages.

Figs. 8 and 9 illustrate the differences among policies at the assignment stage, for parallel and serial systems, respectively. We highlight that the *maximax* k , and *maximax* θ policies seem to perform well generally for the assignment stage. Policy performance at the assignment stage for the parallel system displays results similar to the ones at the selection stage with the top policies being *maximax* policies with regard to O , k and θ . Since the parameter O is obtained by estimating the output of a worker at each task and because specialization is often optimal in parallel systems (see e.g., Nembhard & Bentefouet, 2012), it is expected that a *maximin* assignment policy with regard to O might perform well. The current simulated time horizon allows some workers to reach their steady state productivity, and hence the *maximax* assignment policy based on k is similar in the limit to that for parameter O . In a pure serial system, as illustrated by Bentefouet and Nembhard (2013), the throughput of the production line is the output from the least productive task of the line. The performance of a line depends on all workers assigned to any task on that

line, and hence the gap observed in the parallel system between the best policies and the others is likely to be reduced in the serial system in order to account for these dependencies.

Figs. 10 and 11 illustrate the interaction between the selection and grouping stages, for parallel and serial systems, respectively. These illustrate the relative efficiencies of the selection policies, as well as the fact that the selection stage policies tend to dominate the grouping policies. This is quite clear in the parallel case, in Fig. 10, and true but less distinctly in the serial case, in Fig. 11. Also, in the serial configuration, policies based on different parameters at each stage may have an advantage. For instance, on average the best 2-stage policy, is to use the *maximax* θ selection strategy, and the minimum variance of steady-state productivity, k , grouping strategy.

Considering the pairwise interactions between the Selection and Assignment stages, the results are similar to those for Selection and Grouping, wherein the Selection stage dominates the interaction, particularly in the parallel case. The overall best two-stage policy, is associated with three or fewer selection policies, specifically, *maximax* selection policies with respect to k , θ , and O . Simi-

larly, the pairwise interactions between the Assignment and Grouping stages suggest that the overall interaction is significant in both the parallel and serial cases. However, policy differences do not stand out as sharply, suggesting that the Assignment/Grouping interaction may not be particularly important.

5. Conclusions

This study investigated the operational decision-making processes of selection, grouping, and assignment of workers to tasks, in the context of individual performance variability. Rather than rely on cumbersome nonlinear optimization problems that can be impractical to implement and to solve, we focus on the development of several straightforward policies, which can both provide managerial insight, and potential practical implementation. Parallel and serial task scenarios were examined to represent a range of configurations since many production systems involve a combination of these two types.

In particular we have interest in the grouping stage, where, deciding which workers will work alongside others, may have an

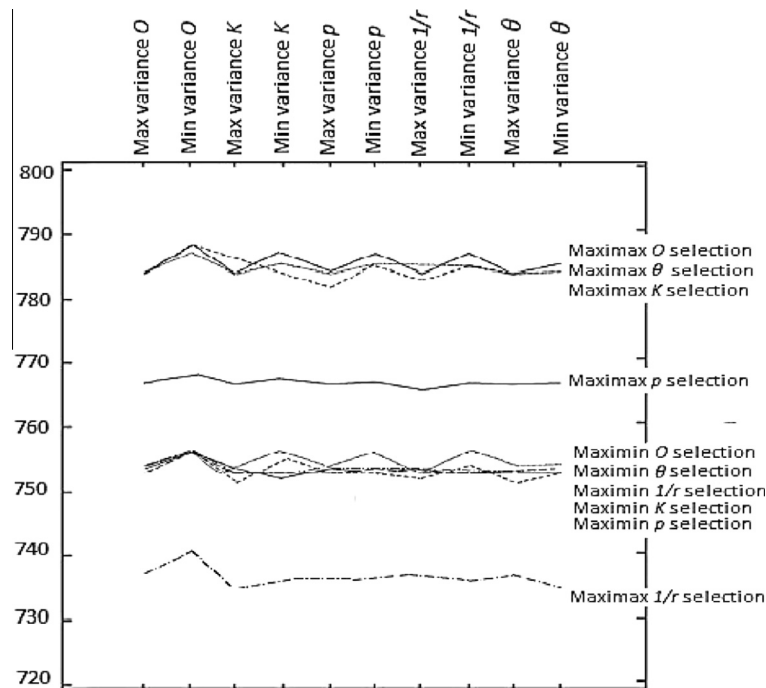


Fig. 10. Interaction: Selection and Grouping – Parallel system.

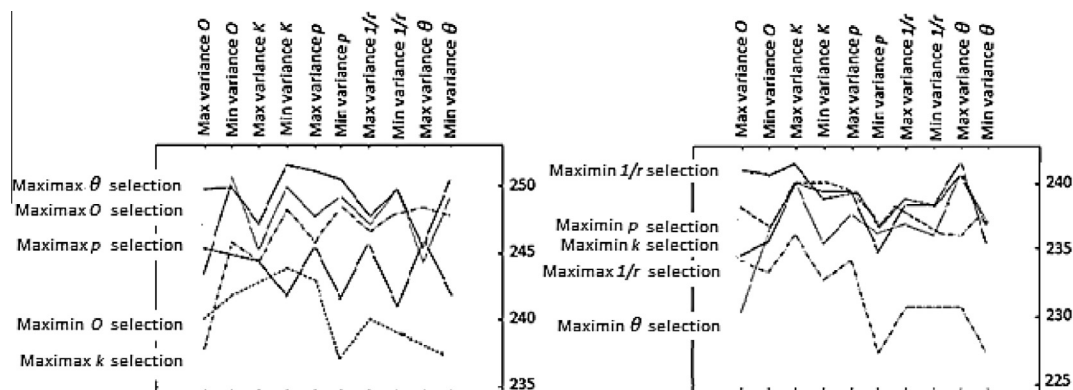


Fig. 11. Interaction: Selection and Grouping – Serial system.

impact on how much transfer of knowledge occurs, in part also dependent on individual transfer parameters. Because we model learning-by-transfer to occur when workers perform similar tasks, the grouping stage consists of assembling workers in teams that are later assigned to a type of task. We extend the original learning model from Mazur and Hastie (1978) to include the transfer of knowledge from group experience to individual performance. The proposed model accounts for learning by doing and learning by transfer of knowledge from co-workers performing similar duties. To distinguish the two phenomena, learning by doing, and learning by transfer, the parameters that account for learning are sampled independently of the parameter that quantifies the transfer of knowledge. Empirical data obtained from a secondary data set from published research is used to generate learning by doing and learning by transfer parameters. We presented top ten heuristic policies for both system types indicating that the selection, grouping, and assignment stages each can be based on different measures. Heuristics based on expected short-term productivity, expected long-run productivity, and knowledge transfer for the selection stage provided the best performance in the parallel task scenario, as the relative gap between the upper bound and the respective heuristics is around 2%.

This study has several managerial implications that may aid managers in the decision stages of selecting, grouping, and assigning workers to tasks, with the overall goal of maximizing production output. We acknowledge that generating optimal solutions can be computationally intensive and thus a challenge to perform for daily operations. Thus, by separating the relevant managerial decisions into three stages, we note insights for practical solutions toward managing productivity. For instance, the worker selection decision has the greatest potential impact on system performance, followed by assignment, and then grouping. Thus, toward prioritizing efforts, management may focus on system decisions in this order. While it is not surprising that selection is important and effective in this regard, we also note that for many organizations, the workforce is often a given, and wholesale changes may not be practical. Nonetheless, for selection, an effective policy is to adopt a greedy (*maximax*) approach for hiring workers based on their individual expected performance. We note that the measure to optimize for selection may depend upon system type, with long run production rate generally preferable for parallel systems, and short term expected production rates more preferable for serial systems.

In organizations with a predominantly *in situ* workforce, task assignment would be the more impactful managerial decision stage, between the two other stages. That is, determining which tasks are to be matched to specific workers would yield the greatest potential improvement when managing an existing workforce. For assignment, greedy policies also appear to perform well. However, the best measures to optimize were long run production rate, and knowledge transfer. We remark that managers should not ignore the degree to which individual workers have capacity for knowledge transfer. While the grouping policy had less of an effect on productivity than selection and assignment, it has a significant effect on performance through the transfer of knowledge from the group's experience to the individual. Managers could note that more homogeneous groups with respect to expected near-term productivity would be expected to perform generally as well as the other policies tested.

We recognize the potential challenges to implement these types of policies. The selection, grouping and assignment processes rely on knowledge of individually measured worker parameters. While it is possible to estimate these parameters from a limited set of production data, we note that many organizations consider previous experience (time) of workers in many of these operational

decisions. However, the current results suggest that prior experience by itself is perhaps not the most useful individual parameter to consider. Among the parameters, the long run steady state productivity parameter would hold that distinction when paired with the appropriate criterion. Given the relevant data, the policies described are readily operationalized, since they do not require complex optimization methods to obtain solutions.

This study is limited to pure parallel and pure serial production systems. We specifically simulated systems with three machines, with three tasks each for with the purpose of prioritizing potentially useful heuristics. While the two production systems studied here form the bases for many production systems, it could be of an interest to extend the proposed methodology to a broader array of specific systems that integrate serial and parallel components explicitly. It may also be of a research interest to assess how the quality and relative usefulness of developed heuristics could relate to the number of tasks and/or machines in the context of much larger systems.

We have identified some additional questions regarding the conditions when knowledge transfer may be beneficial within small work groups. While past research has shown that knowledge transfer is primarily developed through team interactions, transactive memory systems have been shown to develop more effectively between team members when they were trained *together* on the task rather than separately. Moreland and Myaskovsky (2000) provided an alternative to develop the shared store of knowledge between the members by providing team members with feedback on member's skills on the assigned task prior to performing it. Thus, knowledge transfer itself is a complex aspect of the larger production problem. The integration of these additional sources of variability from knowledge transfer mechanisms may yield further insights into managing the workforce.

Historically, schedules have been constructed with the aim of reducing variability in production systems, especially with respect to tasks completion times. Two approaches were widely used: a *task approach*, which stipulates that a major source of variability is the inherent variability of the tasks, and a *workload approach* that claims it is the environment where the tasks are being performed that explains variability. The current study effectively examined a *worker approach*, wherein a major source of variability in task completion times is due to the choice of worker performing the task. While this approach acknowledges worker differences and has been able to explain some of the field observations, the task approach and workload approach continue to be useful in many settings.

Appendix A. Mixed integer nonlinear program for establishing the upper bound

In this section, we present the MINLP formulation that establishes an upper bound to which we evaluate the performance of the implemented heuristics. We denote the formulation for the parallel system Model 1 and the one for the serial system Model 2.

A.1. Model 1: Parallel systems

- j . task index $j = 1, 2, \dots, 9$
- i . worker index $i = 1, 2, \dots, 45$
- t . index of production periods $t = 1, 2, \dots, T$
- L . period duration
- y_{qj} . equals 1 if task q and task j are of the same type and 0 otherwise (is an alias of the task index)
- x_{ijt} . equals 1 if worker i is assigned to task j at time t and 0 otherwise.
- S_{jt} . the output from task j at period t .

P_{ijt} : the productivity rate of worker i assigned to task j at period t .
 O_{ijt} : the output of worker i on task j at time t .

$$\text{Maximize } \sum_{j=1}^9 \sum_{t=1}^T S_{jt}$$

$$S_{jt} = \sum_{i=1}^{45} O_{ijt} \quad j = 1..9 \quad t = 1..T \quad (\text{A-1})$$

$$O_{ijt} = \sum_{k \leq t} x_{ijk} * P_{ijk} \quad i = 1..45 \quad j = 1..9 \quad t = 1..T \quad (\text{A-2})$$

$$T_{ijt} = \sum_{q=1, q \neq j}^9 y_{qj} * \left(\sum_{p=1, p \neq i}^9 \left(\sum_{k \leq t} x_{pqk} * P_{pqk} \right) \right) \quad (\text{A-3})$$

$$P_{ijt} = k_{ij} * \frac{[\theta_{ij} * T_{ijt} + (\sum_{k \leq t} L * x_{ijt}) + p_{ij}]}{[\theta_{ij} * T_{ijt} + (\sum_{k \leq t} L * x_{ijt}) + p_{ij} + r_{ij}]} \quad (\text{A-4})$$

$$\sum_{i=1}^{45} x_{ijt} = 1 \quad j = 1..9 \quad t = 1..T \quad (\text{A-5})$$

$$\sum_{j=1}^9 x_{ijt} \leq 1 \quad i = 1..45 \quad t = 1..T \quad (\text{A-6})$$

Within the parallel production scenario, the objective is to maximize the output from all nine tasks. Constraint (A – 1) defines the throughput at each task j and constraint (A – 2) the units produced by each work at each task. The term T_{ijt} that quantifies the knowledge that can be transferred to a worker from others workers when performing similar tasks, is defined in equation (A – 3). In equation

(A – 4) we define the worker productivity rate for a given task at a given period as a function of the transfer parameter, the learning parameters and the cumulate assignments. Constraint (A – 5) to constraint (A – 6) ensure that at each time t , a task is occupied by exactly one worker and a worker is assigned to exactly task.

A.2. Model 2: Serial systems

In this second scenario, we consider a flow line with three tasks of different types as described in Fig. 2. On each of the three flow lines, a unit must progress through the three different tasks sequentially, and this sequence is constant for all jobs. Thus, the objective is to maximize the number of products that are completed at the final task-station. Bentefouet and Nembhard (2013) establish that the problem, in term of the value of the objective function, is equivalent to the scheduling problem that ignores the buffers constraint and maximizes the least productive tasks. That is, in addition with constraints (A – 1) to, we replace the objective function by:

$$\text{Maximize } \text{Line}_1 + \text{Line}_2 + \text{Line}_3$$

and add the following equations that define the throughput at each of the three lines:

$$\text{Line}_1 = \min \left(\sum_{t=1}^T S_{1t}, \sum_{t=1}^T S_{4t}, \sum_{t=1}^T S_{7t} \right) \quad (\text{A-7})$$

$$\text{Line}_2 = \min \left(\sum_{t=1}^T S_{2t}, \sum_{t=1}^T S_{5t}, \sum_{t=1}^T S_{8t} \right) \quad (\text{A-8})$$

$$\text{Line}_3 = \min \left(\sum_{t=1}^T S_{3t}, \sum_{t=1}^T S_{6t}, \sum_{t=1}^T S_{9t} \right) \quad (\text{A-9})$$

Table 6
Top 30 policies parallel system.

Selection	Grouping	Assignment	Mean throughput	Mean relative gap to the upper bound in % (std in %)
Maximax k	Min variance k	Maximax k	808.89 (3.681)	1.15 (1.03)
Maximax k	Min variance p	Maximax θ	808.89 (3.681)	1.15 (1.03)
Maximax k	Min variance k	Maximax θ	807.77 (3.602)	1.15 (1.06)
Maximax k	Min variance $1/r$	Maximax k	807.77 (3.602)	1.15 (1.06)
Maximax k	Min variance $1/r$	Maximax θ	807.77 (3.602)	1.15 (1.06)
Maximax k	Min variance p	Maximax k	806.76 (4.124)	1.16 (1.06)
Maximax k	Min variance θ	Maximax θ	806.56 (3.818)	1.16 (0.95)
Maximax k	Min variance θ	Maximax k	804.88 (6.148)	1.18 (1.03)
Maximax k	Min variance O	Maximax p	803.02 (14.025)	1.54 (1.36)
Maximax θ	Min variance O	Maximax p	803.00 (13.715)	1.56 (1.43)
Maximax θ	Min variance $1/r$	Maximax k	802.95 (12.356)	1.58 (1.44)
Maximax k	Max variance $1/r$	Maximax θ	802.88 (3.773)	1.64 (1.34)
Maximax θ	Min variance k	Maximax θ	802.42 (13.094)	1.66 (1.41)
Maximax θ	Min variance p	Maximax θ	802.42 (13.094)	1.66 (1.41)
Maximax θ	Min variance $1/r$	Maximax θ	802.42 (13.094)	1.66 (1.41)
Maximax θ	Min variance θ	Maximax θ	802.42 (13.094)	1.66 (1.41)
Maximax θ	Min variance p	Maximax k	802.25 (13.524)	1.69 (1.44)
Maximax θ	Min variance θ	Maximax k	802.25 (13.524)	1.69 (1.44)
Maximax θ	Min variance k	Maximax k	801.73 (13.264)	1.75 (1.55)
Maximax θ	Min variance $1/r$	Maximax O	800.41 (14.652)	1.82 (1.47)
Maximax θ	Min variance O	Maximax θ	800.28 (15.569)	1.83 (1.44)
Maximax k	Min variance O	Maximax θ	800.22 (15.571)	1.84 (1.44)
Maximax k	Min variance k	Maximax O	799.89 (15.810)	1.88 (1.45)
Maximax k	Min variance p	Maximax O	799.89 (15.810)	1.88 (1.45)
Maximax k	Min variance O	Maximax O	799.84 (15.811)	1.89 (1.45)
Maximax k	Min variance $1/r$	Maximax O	799.65 (15.849)	1.91 (1.48)
Maximax k	Min variance θ	Maximax O	799.65 (15.849)	1.91 (1.48)
Maximax k	Min variance O	Maximax k	799.62 (15.850)	1.91 (1.47)
Maximax θ	Min variance k	Maximax O	799.20 (15.495)	1.96 (1.57)
Maximax θ	Min variance O	Maximax O	799.18 (15.495)	1.97 (1.57)

Table 7
Top 30 policies serial system.

Selection	Grouping	Assignment	Mean throughput	Mean relative gap to the upper bound in % (std in %)
Maximax θ	Min variance $1/r$	Maximax O	259.07 (7.670)	6.22 (3.14)
Maximax θ	Min variance k	Maximax θ	258.02 (7.691)	6.28 (3.14)
Maximax θ	Min variance k	Maximax k	258.63 (7.812)	6.38 (3.29)
Maximax k	Min variance O	Maximax O	258.62 (7.409)	6.38 (3.03)
Maximax θ	Max variance p	Maximax k	258.56 (6.421)	6.42 (2.43)
Maximax k	Min variance k	Maximax θ	258.48 (7.275)	6.44 (3.02)
Maximax θ	Max variance p	Maximax θ	258.40 (6.437)	6.47 (2.42)
Maximax k	Min variance p	Maximax θ	258.11 (7.216)	6.57 (2.94)
Maximax O	Min variance k	Maximax k	257.78 (6.994)	6.69 (2.98)
Maximax θ	Min variance O	Maximax $1/r$	257.63 (12.522)	6.75 (4.73)
Maximax O	Min variance p	Maximax θ	257.52 (7.333)	6.78 (3.07)
Maximax k	Min variance k	Maximax k	257.49 (9.257)	6.78 (3.70)
Maximax O	Min variance k	Maximax O	257.29 (7.381)	6.86 (3.07)
Maximax θ	Min variance $1/r$	Maximax k	275.19 (12.74)	6.90 (4.79)
Maximax O	Min variance θ	Maximax O	257.11 (7.226)	6.92 (3.01)
Maximax θ	Min variance k	Maximax O	257.08 (12.851)	6.94 (4.82)
Maximax θ	Min variance O	Maximax k	257.02 (12.851)	6.96 (4.82)
Maximax θ	Min variance O	Maximax θ	257.01 (12.853)	6.97 (4.82)
Maximax θ	Min variance O	Maximax O	256.73 (12.914)	7.07 (4.83)
Maximax O	Min variance p	Maximax O	256.70 (6.903)	7.08 (2.89)
Maximax O	Min variance θ	Maximax p	256.34 (6.764)	7.20 (3.01)
Maximax θ	Max variance $1/r$	Maximax k	256.15 (6.066)	7.28 (2.41)
Maximax θ	Min variance θ	Maximax k	256.07 (12.693)	7.30 (4.86)
Maximax θ	Min variance p	Maximax θ	256.04 (12.399)	7.31 (4.75)
Maximax θ	Min variance p	Maximax k	256.01 (12.135)	7.32 (4.66)
Maximax k	Min variance p	Maximax k	256.00 (12.551)	7.32 (4.68)
Maximax k	Min variance O	Maximax k	255.97 (12.421)	7.34 (4.63)
Maximax k	Min variance $1/r$	Maximax k	255.47 (16.875)	7.52 (6.20)
Maximax k	Min variance k	Maximax O	255.28 (13.815)	7.60 (5.10)
Maximax k	Min variance $1/r$	Maximax θ	255.22 (17.231)	7.62 (6.33)

Appendix B. Top 30 best policies parallel and serial system

(See Tables 6 and 7).

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